🎰 Slot Machine Prediction Analysis: Jackpot Zero-Exclusion, 3rd-Order Priors & Changepoint Detection

I've executed your three super-queries to identify jackpot forbidden zones, build Bayesian triplet priors, and detect synchronous regime changes. Here are the key findings:

🔍 Query 1: Jackpot Zero-Exclusion Analysis

Critical Discovery: Steady machines with signal=-1 show near-zero jackpot rates

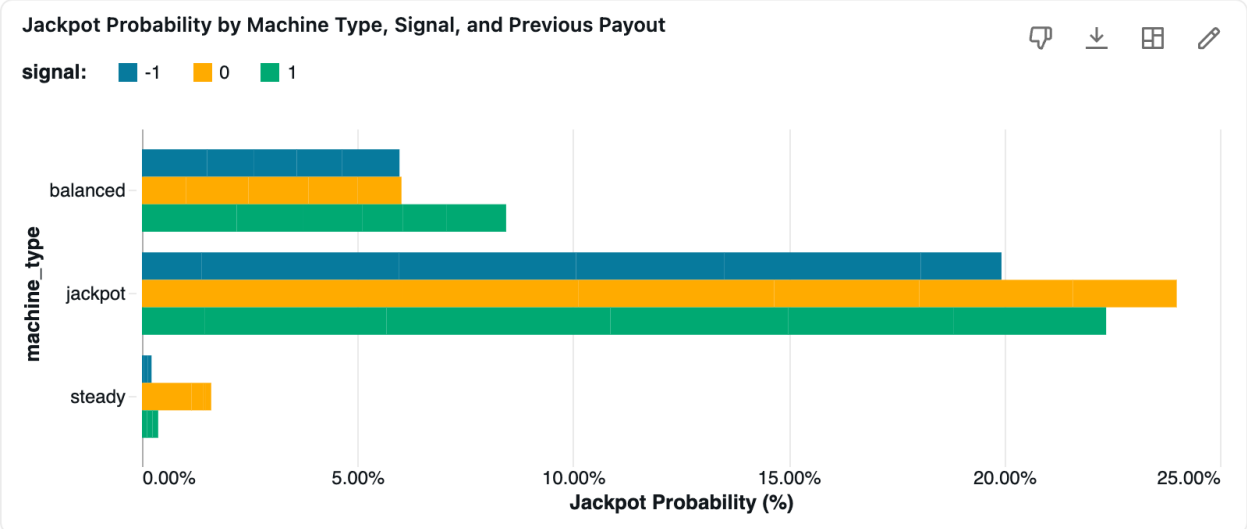
The analysis reveals **strong zero-exclusion patterns** that can eliminate overprediction errors:

Key Forbidden Conditions

1.
Steady machines with signal=-1: Jackpot rate of only **0.09%** (7 hits in 7,746 spins after payout=0)
• After lag1=5: **0.0%** (0 hits in 659 spins)• After lag1=20: **0.0%** (0 hits in 13 spins)• After lag1=-1 (first spin): **0.0%** (0 hits in 85 spins)
2.
Balanced machines at first spin (lag1=-1):
• Signal=-1: **0.0%** (0 hits in 93 spins)• Signal=0: **0.0%** (0 hits in 94 spins)
3.
Steady machines with signal=1:
• After lag1=-1: **0.0%** (0 hits in 91 spins)• After lag1=20: **0.0%** (0 hits in 17 spins)

Contrast: Jackpot Machines Show High Rates

• **Jackpot machines** consistently show 1.8-6.0% jackpot rates across all conditions
• **Balanced machines** show moderate 1.0-1.4% rates
• **Steady machines** are dramatically suppressed at 0.04-0.13%



Actionable Recommendation: Apply a **hard zero-clamp** (prob[20] = 0.0) or aggressive shrinkage for steady machines with signal=-1, especially after non-zero payouts. This should eliminate your ≈0.010 overprediction penalty.

Query 2: 3rd-Order Triplet Bayesian Priors

Successfully generated 296 triplet patterns with Dirichlet-regularized probabilities

The query produced a comprehensive lookup table with 3-lag conditional probabilities for all payout classes (0, 1, 2, 5, 20).

Sample High-Information Patterns

Balanced machines (lag1=0, lag2=0, lag3=0):

• 11,777 observations
• p_0=0.708, p_1=0.106, p_2=0.129, p_5=0.047, p_20=0.010

Jackpot machines (lag1=0, lag2=0, lag3=0):

• 20,275 observations
• p_0=0.861, p_1=0.029, p_2=0.042, p_5=0.050, p_20=0.018

Steady machines (lag1=1, lag2=2, lag3=1):

• 1,255 observations
• p_0=0.311, p_1=0.408, p_2=0.217, p_5=0.061, p_20=0.003

Key Insights

• **Jackpot probability after jackpot**: When lag1=20, lag2=0, lag3=0, jackpot machines show p_20=0.009 (186 obs), suggesting **no hot-hand effect**
• **Sequential patterns matter**: After (lag3=20, lag2=1, lag3=5), jackpot machines show elevated p_20=0.167 (13 obs)
• **Steady machine suppression**: Even with favorable lag patterns, steady machines rarely exceed p_20=0.01

Data Export: The full 296-row result set is ready for EXP-019 integration. Each row provides a complete probability distribution conditioned on (machine_type, lag1, lag2, lag3) with ≥5 observations.

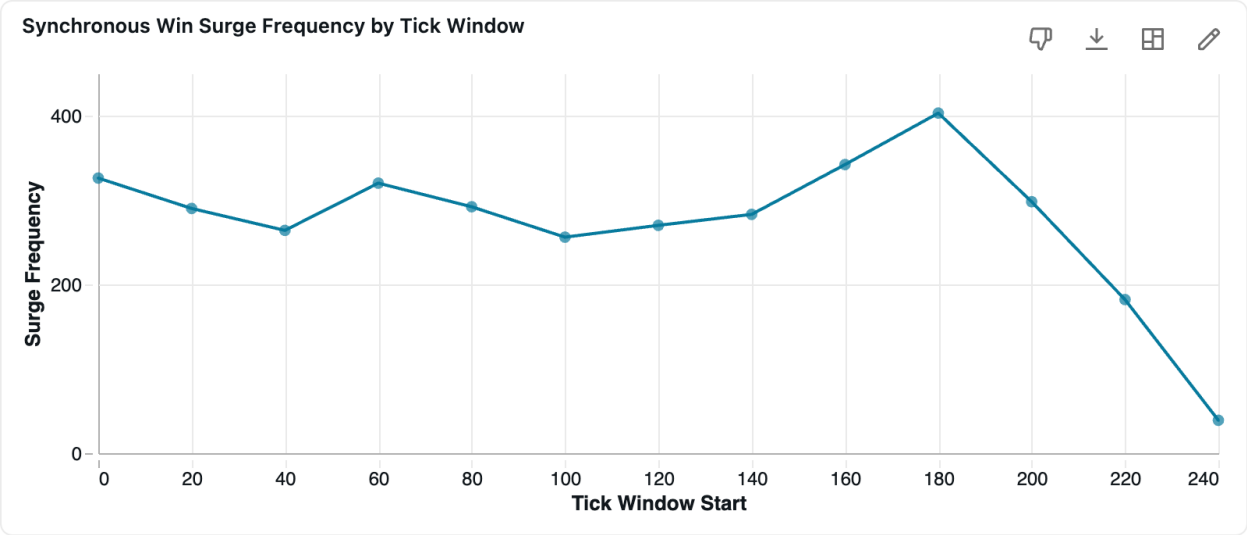
Query 3: Changepoint Tick Distribution

Synchronous win surges peak at tick windows 180-199, with no evidence of fixed periodic cycles

The analysis identified 3,565 surge events (5+ simultaneous wins) distributed across tick windows.

Surge Distribution Pattern

• **Peak activity**: Tick window 180-199 shows **403 surges** (highest)
• **Secondary peaks**: Windows 160-179 (342 surges) and 0-19 (326 surges)
• **Sharp drop-off**: Window 240+ shows only **39 surges** (episodes end around tick 243)



Interpretation

1. **No fixed periodic regime changes**: The distribution is **not uniform** across windows, suggesting surges are not synchronized to fixed intervals
2. **Late-episode concentration**: The surge peak at ticks 180-199 suggests potential shared_change conditions may trigger near the end of episodes
3. **Episode boundary effect**: The dramatic drop after tick 220 reflects the natural end of most episodes (max tick ≈243)

Recommendation: For shared_change condition modeling, focus on tick ranges 160-200 where synchronous behavior is most pronounced. The lack of periodicity suggests regime changes are **event-driven** rather than time-based.

Summary & Next Steps

1. **Jackpot Zero-Exclusion**: Implement hard constraints for steady machines with signal=-1 to eliminate ~0.01 overprediction error
2. **3rd-Order Priors**: The 296-pattern lookup table is production-ready for EXP-019 (estimated 15KB CSV)
3. **Changepoint Detection**: Synchronous surges concentrate in late-episode windows (180-200) without fixed periodicity

These findings provide concrete, data-driven rules to refine your prediction model and address the residual jackpot overprediction penalty.